Max Ellis *BDes MSc AFHEA*

PhD Candidate in Psychoacoustics

Acoustics Innovation Institute, University of Salford
EPSRC Centre for Doctoral Training in Sustainable Sound Futures

ellisacoustics.github.io · linkedin.com/in/psychoacoustic · github.com/ellisacoustics · orcid.org/0009-0007-5909-854X

Drones, air taxis and heat pumps are about to change how our towns and cities sound forever. I work out how people will perceive them — *before* they arrive.

Noise already causes more ill health in Europe than any environmental factor except air pollution[WHO, 2011]. The next wave of technology will introduce noise that loudness alone cannot accurately judge[House of Lords, 2023]: the high-pitched whine of a rotor, the tonal hum of a compressor, rumbles of large aircraft flying low over gardens. My PhD research, sponsored by the Department for Environment, Food and Rural Affairs (Defra), seeks to accurately simulate these future soundscapes to gather response data from listening panels, for machine learning models to predict who will be annoyed, to what extent, and why. That gives regulators evidence to set standards on, and engineers a target for quieter machines, before they are everywhere.

Research Collaborations

Defra — PhD Sponsor
HEAD Acoustics GmbH — Journal Paper
Joby Aviation Ltd. — Journal Paper

Publications

2027

Multi-Model Analysis for Prediction of Perceived Annoyance from UAVs

M. W. Ellis, T. Lobato, T. Kamper-Schley, R. Sottek, M. C. Green, Z. Podwińska, A. J. Torija
Journal of the Acoustical Society of America (JASA), 2027.

Pending | Journal Paper

2027

Psychoacoustic Assessment of UAV Noise in Varying Ambiences, Operational Conditions and Contexts

M. W. Ellis, Z. Podwińska, A. J. Torija
Forum Acusticum 2027, Brussels.

Awarded Best Student Paper

`Pending` `Conference Paper`

2027

Impacts on Soundscape Perception from eVTOL Flyovers in Existing Urban Ambiences

M. W. Ellis, N. Shrubsole-Green, J. Bain, Z. Podwińska, A. J. Torija
AIAA 2027, San Diego.

`Pending` `Conference Paper`

2026

Comparing a General Sound Quality Metric for Aircraft Noise Using a Pre-Trained Neural Network Model to Traditional Psychoacoustics

T. Lobato, T. Kamper-Schley, R. Sottek, M. C. Green, **M. W. Ellis**, A. J. Torija
Quiet Drones 2026, Delft. Human Response to UAS and UAM Noise. With HEAD Acoustics GmbH.

PDF `Conference Paper`

2025

Comparison of Deep Learning and Psychoacoustic Models to Predict UAV Noise Impact in Soundscapes

M. W. Ellis, M. C. Green, M. J. B. Lotinga, A. J. Torija Martínez
Forum Acusticum EuroNoise 2025, 11th Convention of the European Acoustics Association, Málaga, pp. 2121–2127.

doi:10.61782/fa.2025.0235 PDF

`Conference Paper` `Open Access`

All publications on Google Scholar · ORCID

Software Proficiencies

Python R MATLAB Max/MSP ArtemiS SUITE QGIS FME REAPER

Experience

Aug 2027 – Oct 2027

Research Engineer (PhD Placement)

Dyson

- Assessing whether structural vibration can underpin sound quality metrics for product components, comparing vibration and sound quality data across anechoic, semi-reverberant and reverberant conditions.
- Applying transfer learning across sound quality metrics and jury test data from other products, to build machine learning models that predict listener response and satisfaction.

`Machine Learning` `Psychoacoustics` `Jury Testing` `DSP` `Acoustical Engineering`

Sep 2026 – May 2027

Visiting Lecturer

University of the West of Scotland

- 1st Supervisor on the Major Project (Dissertation) module for BSc (Hons) Music Technology students specialising in Acoustics and Environmental Noise.

`University Teaching`

Aug 2026 – Oct 2026

Specialist Consultant (PhD Placement)

Noise Consultants Ltd. (part of Logika Group)

- Validated open-source noise measurements against modelled Defra noise-mapping data for infrastructure assessment.
- Ran a field campaign on an airport expansion project comparing physical and perceptual dominance of aircraft noise.

`Field Measurements` `Statistical Modelling` `Data Visualisation` `Acoustical Engineering`

2026 – Present

Associate Lecturer, Computer Science

St Mary’s University, Twickenham

- Teaching, marking and moderation across the online MSc Computer Science suite.
- Module reviewer and author.

`University Teaching`

2026 – Present

Teaching Assistant, Applied Acoustics

University of Salford

- Lab demonstrator for Environmental Noise Measurement and Modelling (BEng/MSc).
- Tutor for Immersive Sound Reproduction (MSc/PgDip).

University Teaching | Field Measurements | Spatial Audio

Education

2025 – 2028

PhD Acoustics and Audio Engineering

THESIS

Psychoacoustic Assessment and Machine Learning-Based Modelling of Novel Noise Source Perception in Complex Soundscapes

University of Salford Acoustics Innovation Institute

Psychoacoustics | Auralisation | Jury Testing | Machine Learning | Spatial Audio | DSP | Statistical Modelling | Data Visualisation

2023 – 2024

MSc Computer Science with Artificial Intelligence

DISSERTATION

A Future Research Framework: Psychoacoustic Annoyance Models and Deep Learning for Prediction of UAV Noise Affect

Distinction

Abertay University

Machine Learning | Data Visualisation

2021 – 2023

BDes (Hons) Sound for Moving Image

DISSERTATION

A Comparative Analysis of AI Tools for Electronic Music Production: Large Language Models, Generative-AI and Virtual Composers

2:1

University of Glasgow

Spatial Audio | Field Measurements

2019 – 2021

DipHE Music Technology

Distinction

University of the West of Scotland

Awarded BSc Level 8 Court Medal

DSP | Acoustical Engineering

2019

BTEC Level 3 Digital Synthesis and Sampling

Pass

Fife College

2018 – 2019

SCQF Level 6 Sound Production

Pass

Glasgow Clyde College Anniesland

Working Groups and Committees

NASA Unmanned Aircraft Systems Noise Working Group (UNWG) — Member
Noise Network Plus A.I. and Digital Working Group — Member
Institute of Acoustics Scottish Branch and Early Careers Group — Committee Member

Certifications

Advanced Prompt Engineering
Vanderbilt University

Advanced R Programming
Johns Hopkins University

AI for Noise Pollution Control
LSIB

Applications of AI in Acoustics
European Acoustics Association Summer School

Auralization
Autumn School Series in Acoustics (ASSA) 2026

Digital Signal Processing
EPFL

Ethics of Artificial Intelligence
Politecnico di Milano

Responsible Research and Innovation
ORBITRRI

Transformer Models and BERT Model
Google

Trustworthy Generative AI
Vanderbilt University